

Class 10 Mathematics Formula Sheet

(CBSE / ICSE / State Boards)

1. Real Numbers

Euclid's Division Lemma:

For integers $a, b > 0$, $\exists$ unique q, r such that

$$a = bq + r, \quad 0 \leq r < b$$

Relation between HCF and LCM:

$$\text{HCF}(a, b) \cdot \text{LCM}(a, b) = a \cdot b$$

2. Polynomials

Quadratic Polynomial: For $ax^2 + bx + c$ with roots α, β :

$$\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}$$

Cubic Polynomial: For $ax^3 + bx^2 + cx + d$ with roots α, β, γ :

$$\alpha + \beta + \gamma = -\frac{b}{a}, \quad \alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}, \quad \alpha\beta\gamma = -\frac{d}{a}$$

3. Pair of Linear Equations

General form:

$$a_1x + b_1y + c_1 = 0, \quad a_2x + b_2y + c_2 = 0$$

Conditions:

$$\begin{array}{ll} \text{Unique solution:} & \frac{a_1}{a_2} \neq \frac{b_1}{b_2} \\ \text{No solution:} & \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \\ \text{Infinite solutions:} & \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \end{array}$$

Cross-multiplication method:

$$\frac{x}{b_1c_2 - b_2c_1} = \frac{y}{c_1a_2 - c_2a_1} = \frac{1}{a_1b_2 - a_2b_1}$$

4. Quadratic Equations

Standard form: $ax^2 + bx + c = 0$, $a \neq 0$

Discriminant: $D = b^2 - 4ac$

Nature of roots:

$D > 0$ Two distinct real roots

$D = 0$ Equal real roots

$D < 0$ No real roots

Quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

5. Arithmetic Progressions

n -th term:

$$a_n = a + (n - 1)d$$

Sum of first n terms:

$$S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$$

6. Coordinate Geometry

Distance:

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Section formula:

$$\left(\frac{mx_2 + nx_1}{m + n}, \frac{my_2 + ny_1}{m + n} \right)$$

Midpoint:

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Area of triangle:

$$\frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

7. Triangles

Basic Proportionality Theorem:

$$\frac{AD}{DB} = \frac{AE}{EC}$$

Pythagoras Theorem:

$$a^2 + b^2 = c^2$$

Similarity:

$$\frac{AB}{PQ} = \frac{BC}{QR} = \frac{CA}{RP}$$

8. Trigonometry

Ratios:

$$\sin \theta = \frac{\text{opp}}{\text{hyp}}, \quad \cos \theta = \frac{\text{adj}}{\text{hyp}}, \quad \tan \theta = \frac{\text{opp}}{\text{adj}}$$

Reciprocals:

$$\csc \theta = \frac{1}{\sin \theta}, \quad \sec \theta = \frac{1}{\cos \theta}, \quad \cot \theta = \frac{1}{\tan \theta}$$

Identities:

$$\sin^2 \theta + \cos^2 \theta = 1, \quad 1 + \tan^2 \theta = \sec^2 \theta, \quad 1 + \cot^2 \theta = \csc^2 \theta$$

9. Circles

- Radius $\perp$ tangent at contact
- Tangents from external point equal
- Tangent-secant theorem: $PT^2 = PA \cdot PB$

10. Areas Related to Circles

$$\text{Area} = \pi r^2, \quad \text{Circumference} = 2\pi r$$

$$\text{Arc length} = \frac{\theta}{360} \cdot 2\pi r, \quad \text{Sector area} = \frac{\theta}{360} \cdot \pi r^2$$

11. Surface Areas and Volumes

Shape	CSA	TSA	Volume
Cuboid (l, b, h)	$2h(l + b)$	$2(lb + bh + hl)$	lbh
Cube (a)	$4a^2$	$6a^2$	a^3
Cylinder (r, h)	$2\pi rh$	$2\pi r(r + h)$	$\pi r^2 h$
Cone (r, h, l)	πrl	$\pi r(l + r)$	$\frac{1}{3}\pi r^2 h$
Sphere (r)	—	$4\pi r^2$	$\frac{4}{3}\pi r^3$
Hemisphere (r)	$2\pi r^2$	$3\pi r^2$	$\frac{2}{3}\pi r^3$

12. Statistics

Mean (direct):

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$$

Median:

$$M = L + \left(\frac{\frac{N}{2} - cf}{f} \right) \cdot h$$

Mode:

$$\text{Mode} = L + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \cdot h$$

13. Probability

$$P(E) = \frac{\text{Favourable outcomes}}{\text{Total outcomes}}, \quad 0 \leq P(E) \leq 1$$

$$P(E) + P(\overline{E}) = 1$$